Stochastic Contextual Bandits / Part 1

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Download at https://github.com/MargauxBregere/bandits_for_electricity_systems

K–armed stochastic bandits

Simplest possible framework

K probability distributions $\nu_1, \ldots, \nu_K$ in a model $\mathcal{D}$
with expectations $\mu_1, \ldots, \mu_K$ $\longrightarrow$ $\mu^\star = \max_{a \in [K]} \mu_a$

At each round $t = 1, 2, \ldots,$

1. Statistician picks arm $A_t \in [K]$ (possibly at random)
2. She gets a reward Y_t drawn according to ν_{A_t}
3. This is the only feedback she receives

$\longrightarrow$ Exploration–exploitation dilemma
estimate the ν_a vs. get high rewards Y_t

Goal:
Maximize expected cumulative rewards $\longleftrightarrow$ Minimize regret

$$R_T = T\mu^\star - \mathbb{E}\left[\sum_{t=1}^{T} Y_t\right] = \sum_{a \in [K]} \left(\mu^\star - \mu_a\right) \mathbb{E}\big[N_a(T)\big]$$

$\longleftrightarrow$ Sub-linear control of the $\mathbb{E}\big[N_a(T)\big]$ for suboptimal arms a

Motivation(s)

One-armed bandits in casinos $\rightarrow$ K–armed stochastic bandits

Clinical trials? (cf. K treatments)

Online advertisement (cf. K ads and homogeneous population)

Optimization of a noisy environment

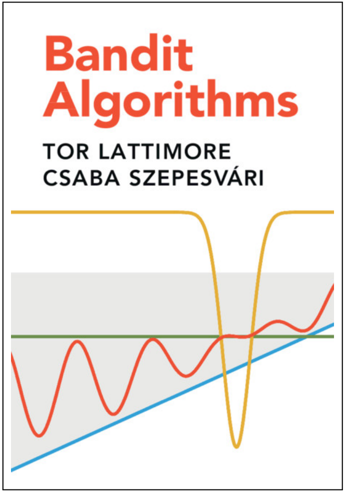
– Continuum of arms $\leftrightarrow$ optimization of a (regular) function
– Tree structure $\leftrightarrow$ chess, go

These lectures are targeted toward a specific application:
Balance between demand and production in electricity systems

I.e., Brégère, Gaillard, Goude, Stoltz, ICML 2019

We will cover what is minimally required to do so

Excellent reference



Bandit Algorithms, Tor Lattimore and Csaba Szepesvári

Cambridge University Press, 2020

Setting of K–armed stochastic bandits:
Distributions $\nu_1, \ldots, \nu_K$ in model $\mathcal{D}$ with expectations $\mu_1, \ldots, \mu_K$
At each round $t \geqslant 1$, pick arm $A_t \in [K]$, get and observe $Y_t \sim \nu_{A_t}$

Proof of the rewriting of regret

Tower rule: $\mathbb{E}[Y_t \mid A_t] = \mu_{A_t}$ thus $\mathbb{E}[Y_t] = \mathbb{E}[\mu_{A_t}]$

$$
\begin{aligned}
R_T &= \sum_{t=1}^{T} \bigl(\mu^\star - \mathbb{E}[Y_t]\bigr) = \sum_{t=1}^{T} \bigl(\mu^\star - \mathbb{E}[\mu_{A_t}]\bigr) \\
&= \sum_{t=1}^{T} \sum_{a \in [K]} \bigl(\mu^\star - \mu_a\bigr)\, \mathbb{E}\bigl[\mathbb{I}_{\{A_t = a\}}\bigr] = \sum_{a \in [K]} \bigl(\mu^\star - \mu_a\bigr)\, \mathbb{E}\bigl[N_a(T)\bigr]
\end{aligned}
$$

where $N_a(T) = \sum_{t=1}^{T} \mathbb{I}_{\{A_t = a\}}$

Noise assumption: σ^2–sub-Gaussian

$$Y_t \sim \nu_{A_t} \qquad \text{made more precise:} \qquad Y_t = \mu_{A_t} + \eta_t$$

Noise η_t independent from "everything else" and σ^2–sub-Gaussian:

$$\forall \lambda \in \mathbb{R}, \qquad \mathbb{E}\big[\mathrm{e}^{\lambda \eta_t} \mid \mathcal{F}_{t-1}\big] = \mathbb{E}\big[\mathrm{e}^{\lambda \eta_t}\big] \leqslant \mathrm{e}^{\lambda^2 \sigma^2/2}$$

$\mathcal{F}_{t-1}$ represents the information: $Y_1, \ldots, Y_{t-1}$ and $A_1, \ldots, A_t$

In particular, $\mathbb{E}\big[\mathrm{e}^{\lambda \eta_t} \mid \mathcal{F}_{t-1}\big] = \mathbb{E}\big[\mathrm{e}^{\lambda \eta_t}\big] = 0$ [proof by Taylor expansions]

Examples:
- $\mathcal{N}(0, \sigma^2)$ noise
- noise taking values in $[a, b]$ is $(b-a)^2/4$–sub-Gaussian (cf. Hoeffding's lemma)

Explore-then-commit strategy

As a warm-up

Explore-then-commit strategy

Reference: e.g., Lattimore and Szepesvári [2020, Chapter 6]

1. Explore for mK rounds

Pull each arm m times, compute empirical means $\widehat{\mu}_a$ for $a \in [K]$

2. Exploit for $T - mK$ rounds

For $t \geqslant mK + 1$, pull an arm $\quad A_t \in \arg\max_{a \in [K]} \widehat{\mu}_a$

Regret bound (under the σ^2–sub-Gaussian noise assumption):

$$R_T \leqslant \underbrace{m \sum_{a \in [K]} (\mu^\star - \mu_a)}_{\text{exploration cost}} + \underbrace{(T - mK) \sum_{a \in [K]} (\mu^\star - \mu_a) \exp\left(-\frac{m(\mu^\star - \mu_a)^2}{4\sigma^2}\right)}_{\text{exploitation cost}}$$

Hoeffding's inequality

Reminder: η is σ^2–sG if $\mathbb{E}\left[e^{\lambda\eta}\right] \leqslant e^{\lambda^2\sigma^2/2}$

If η, η' are σ^2–sG and independent, then $(\eta+\eta')/2$ is $\sigma^2/2$–sG

Cramér-Chernoff: for $\varepsilon > 0$, for $\lambda = \varepsilon/\sigma^2 > 0$,

$$\mathbb{P}(\eta \geqslant \varepsilon) \leqslant e^{-\lambda\varepsilon}\,\mathbb{E}\left[e^{\lambda\eta}\right] \leqslant \exp\left(-\lambda\varepsilon + \lambda^2\sigma^2/2\right) = \exp\left(-\varepsilon^2/(2\sigma^2)\right)$$

Hoeffding's inequality: $\eta_1, \ldots, \eta_m$ independent and σ^2–sG

$$\mathbb{P}\left(\frac{1}{m}\sum_{t=1}^{m}\eta_t \geqslant \varepsilon\right) \leqslant \exp\left(-m\,\varepsilon^2/(2\sigma^2)\right)$$

or w.p. $1-\delta$, $$\frac{1}{m}\sum_{t=1}^{m}\eta_t < \sigma\sqrt{\frac{2}{m}\ln(1/\delta)}$$

Back to explore-then-commit

Regret bound (with σ^2–sub-Gaussian noises):

$$R_T = \sum_{t=1}^{T} (\mu^\star - \mathbb{E}[\mu_{A_t}]) \leqslant \underbrace{m \sum_{a\in[K]} (\mu^\star - \mu_a)}_{\text{exploration cost}} + \underbrace{(T - mK) \sum_{a\in[K]} (\mu^\star - \mu_a) \exp\left(-\frac{m(\mu^\star - \mu_a)^2}{4\sigma^2}\right)}_{\text{exploitation cost}}$$

Exploitation costs $\longleftrightarrow$ $a^\star \in \arg\max_{a\in[K]} \mu_a \neq \arg\max_{a\in[K]} \widehat{\mu}_a$

$$\mathbb{P}\big(\widehat{\mu}_a \geqslant \widehat{\mu}_{a^\star}\big) = \mathbb{P}\left(\frac{1}{m}\sum_{t=1}^{m} \underbrace{\eta'_t - \eta''_t}_{2\sigma^2\text{-sG}} \geqslant \mu^\star - \mu_a\right) \leqslant \big(-m\,\varepsilon^2/(4\sigma^2)\big)$$

where $\varepsilon = \mu^\star - \mu_a$

Two types of regret bounds

Gap-dependent regret bound:

$$R_T = \sum_{t=1}^{T} (\mu^\star - \mathbb{E}[\mu_{A_t}]) \leqslant \underbrace{m \sum_{a \in [K]} (\mu^\star - \mu_a)}_{\text{exploration cost}} + \underbrace{(T - mK) \sum_{a \in [K]} (\mu^\star - \mu_a) \exp\left(- \frac{m(\mu^\star - \mu_a)^2}{4\sigma^2} \right)}_{\text{exploitation cost}}$$

Involves the gaps $\Delta_a = \mu^\star - \mu_a$

More generally called: problem-dependent bound

Now, uniform bound (gap-free bound or worst-case bound):

in a model $\mathcal{D}_{\sigma^2, B}$ with $\Delta_a \leqslant B$

Using $xe^{-x^2} \leqslant 1/2$ for $x \geqslant 0$

$$\sup_{\mathcal{D}_{\sigma^2, B}} R_T \leqslant m\, KB + (T - mK) \frac{2\sigma}{\sqrt{m}} \frac{1}{2} \leqslant m\, KB + \sigma \frac{T}{\sqrt{m}} = \mathcal{O}\left(T^{2/3}\right)$$

for $m \sim T^{2/3}$

Upper-confidence-bound [UCB] strategy

"Optimism in the face of uncertainty"

Model: $\nu_1, \ldots, \nu_K$ are σ^2–sG

A popular strategy: UCB [upper confidence bound]

Auer, Cesa-Bianchi and Fisher [2002]

Pull each arm once

For $t \geqslant K$, pick $$A_{t+1} \in \operatorname*{arg\,max}_{a \in [K]} \left\{ \widehat{\mu}_a(t) + \sigma \sqrt{\frac{8 \ln t}{N_a(t)}} \right\}$$

Exploitation: cf. empirical mean $\widehat{\mu}_a(t) = \dfrac{1}{N_a(t)} \displaystyle\sum_{s=1}^{t} Y_s \, \mathbb{I}_{\{A_s = a\}}$

Exploration: cf. $\sigma \sqrt{8 \ln t / N_a(t)}$ favors arms a not pulled often

Regret bounds (slightly suboptimal) of two types

– Gap-dependent bound: $$R_T \lesssim \square \sum_{a : \mu_a < \mu^\star} \frac{\sigma \ln T}{\mu^\star - \mu_a}$$

– Gap-free bound: $$\sup_{\sigma^2\text{-sG}} R_T \lesssim \square\, \sigma \sqrt{KT \ln T}$$

Proof of $R_T \lesssim \square \sum_{a:\mu_a<\mu^\star} \frac{\sigma \ln T}{\mu^\star - \mu_a}$

Hoeffding–Azuma ext.: $\mathbb{P}\left(\left|\mu_a - \widehat{\mu}_a(t)\right| \leqslant \sigma\sqrt{\frac{8\ln t}{N_a(t)}}\right) \geqslant 1 - 2t^{-3}$

Recall $\quad \widehat{\mu}_a(t) = \frac{1}{N_a(t)} \sum_{s=1}^{t} Y_s\, \mathbb{I}_{\{A_s=a\}}$

Indeed, by Doob's optional skipping:

$$
\begin{aligned}
&\mathbb{P}\left(\left|\mu_a - \widehat{\mu}_a(t)\right| > \sigma\sqrt{\frac{8\ln t}{N_a(t)}}\right) \\
&= \sum_{n=1}^{t} \mathbb{P}\left(\left|\mu_a - \widehat{\mu}_{a,n}\right| > \sigma\sqrt{\frac{8\ln t}{n}} \quad \text{and} \quad N_a(t) = n\right) \\
&\leqslant \sum_{n=1}^{t} \underbrace{\mathbb{P}\left(\left|\mu_a - \widehat{\mu}_{a,n}\right| > \sigma\sqrt{\frac{2\ln(1/t^{-4})}{n}}\right)}_{\leqslant 2t^{-4}, \quad \text{cf. classic Hoeffding-Azuma applied twice}}
\end{aligned}
$$

where $\widehat{\mu}_{a,n}$ denotes a n–sample average with distribution ν_a

Proof of $R_T \lesssim \square \sum_{a:\mu_a<\mu^\star} \frac{\sigma \ln T}{\mu^\star - \mu_a}$

Hoeffding–Azuma ext.: $\mathbb{P}\left(\left|\mu_a - \widehat{\mu}_a(t)\right| \leqslant \sigma \sqrt{\frac{8 \ln t}{N_a(t)}} \right) \geqslant 1 - 2t^{-3}$

If $A_t = b$ is not an optimal arm $a^\star$, then by design

$$\widehat{\mu}_{a^\star}(t) + \sigma \sqrt{\frac{8 \ln t}{N_{a^\star}(t)}} \leqslant \widehat{\mu}_b(t) + \sigma \sqrt{\frac{8 \ln t}{N_b(t)}}$$

thus w.p. $\geqslant 1 - 4t^{-3}$,

$$\mu^\star \leqslant \mu_b + 2\sigma \sqrt{\frac{8 \ln t}{N_b(t)}}$$

which imposes

$$N_b(t) \leqslant \frac{32\sigma \ln T}{(\mu^\star - \mu_b)^2}$$

$\mathbb{E}\big[N_b(T)\big]$ controlled by this $+ \square \sum_{t=K+1}^{T} t^{-2} \leqslant \square$

Conclude with

$$R_T = \sum_{a\in[K]} (\mu^\star - \mu_a)\, \mathbb{E}\big[N_a(T)\big]$$

Proof of $\sup_{\sigma^2\text{-sG}} R_T \lesssim \sigma\sqrt{KT \ln T}$

We proved $$\mathbb{E}\big[N_b(t)\big] \lesssim \square\,\sigma \frac{\ln T}{(\mu^\star - \mu_b)^2}$$

Thus $$R_T = \sum_{a\in[K]} (\mu^\star - \mu_a)\sqrt{\mathbb{E}\big[N_a(T)\big]}\sqrt{\mathbb{E}\big[N_a(T)\big]}$$

$$\leqslant \square\,\sigma\sqrt{\ln T}\sum_{a\in[K]}\sqrt{\mathbb{E}\big[N_a(T)\big]}$$

$$\leqslant \square\,\sigma\sqrt{KT\ln T}$$ cf. Jensen for $\sqrt{}$

Thompson sampling

A Bayesian view

Thompson sampling: a Bayesian view

Studied mostly for exponential families $\mathcal{D} \leftrightarrow \Theta$

E.g., Bernoulli $\Theta = (0,1)$; Gaussian $\Theta = \mathbb{R}$ with known variance σ^2

Ref.: Thompson [1933]; Agrawal and Goyal [2012, 2013, 2017];
Kaufmann, Korda and Munos [2012, 2013]

Well-chosen prior distribution π over Θ

$\mathcal{D}$ Bernoulli $\rightarrow \pi$ Beta; $\mathcal{D}$ Gaussian $\rightarrow \pi$ Gaussian

Posterior distributions: $\widehat{\pi}_{t,a}$ for arm $a \in [K]$

Based on π and rewards Y_s such that $1 \leqslant s \leqslant t-1$ and $A_s = a$

Strategy: draw $\overbrace{\theta_{t,a} \sim \widehat{\pi}_{t,a}}^{\text{exploration}}$ and pick $\overbrace{A_t \in \operatorname*{arg\,max}_{a \in [K]} \mu(\theta_{t,a})}^{\text{exploitation}}$

Thompson sampling: Bernoulli case $\Theta = (0,1)$

Distributions indexed by $p_1, \ldots, p_K$ with best parameter $p^\star$

Prior $\pi = \mathrm{Beta}(1,1) =$ uniform distribution over $(0,1)$

Posteriors: $$\widehat{\pi}_{t,a} = \mathrm{Beta}\big(1 + S_a(t),\ 1 + N_a(t) - S_a(t)\big)$$

where $$N_a(t) = \sum_{s \leqslant t} \mathbb{I}_{\{A_s = a\}}$$ and $$S_a(t) = \sum_{s \leqslant t} Y_s \, \mathbb{I}_{\{A_s = a\}}$$

Regret bounds (complex proofs)

$$R_T \lesssim \sum_{a : p_a < p^\star} (p^\star - p_a) \frac{\ln T}{\mathrm{kl}(p_a, p^\star)} \leqslant \sum_{a : p_a < p^\star} \frac{\ln T}{2(p^\star - p_a)}$$

and also $$R_T \lesssim \square \sqrt{KT \ln T}$$

Optimal regret bounds

Achieved with more complex strategies

Optimal distribution-dependent bound

For all reasonable strategies, $\forall \nu_1, \ldots, \nu_K$ in $\mathcal{D}$,

$$\liminf \frac{R_T}{\ln T} \geqslant \sum_{k \in [K]} \frac{\mu^\star - \mu_k}{\mathcal{K}_{\inf}(\nu_k, \mu^\star)} \qquad \text{where}$$

$$\mathcal{K}_{\inf}(\nu_k, \mu^\star) = \inf\Bigl\{ \mathrm{KL}(\nu_k, \xi) : \ \xi \in \mathcal{D}, \quad \nu_k \ll \xi \ \text{ and } \ \mathrm{E}(\xi) > \mu^\star \Bigr\}$$

References: Lai et Robbins '85, Burnetas et Katehakis '96, Ménard et al. '19

Note: "$\forall \nu_1, \ldots, \nu_K$ in $\mathcal{D}$" after "for all [...] strategies"

Bound achieved by IMED and KL-UCB strategies:
models $\mathcal{D}$ given by 1-dim. exponential families or by $\mathcal{P}\bigl([0,1]\bigr)$

IMED: Honda and Takemura, series of papers in the 2010s

kl-UCB and KL-UCB: Garivier, Cappé, Ménard, Maillard, Munos, Hadiji and Stoltz, series of papers in the 2010s

Optimal distribution-free bound for models

$$\mathcal{D} \supseteq \mathcal{B}er = \big\{ \mathrm{Ber}(p) : p \in (0,1) \big\}$$

For all strategies,

$$\sup\Big\{ R_T : \nu_1, \ldots, \nu_K \text{ in } \mathcal{B}er \Big\} \geqslant \Box\sqrt{TK}$$

References: Auer, Cesa-Bianchi, Freund, Schapire 2002

Bound achieved over $\mathcal{D} = \mathcal{P}\big([0,1]\big)$ and even in the adversarial setting by MOSS

Audibert and Bubeck, 2009; also Tsallis-INF strategies, Zimmert and Seldin, 2018

kl-UCB+ and KL-UCB-switch strategies achieve simultaneously optimal distribution-dependent and distribution-free regret bounds

Still based on upper confidence bounds; Ménard and Garivier, 2017; Garivier, Hadiji, Ménard, Stoltz, 2022

$\leftarrow$ It's all about finer / more efficient upper-confidence bounds!